

DV90 scATAC-seq CNA Caller Benchmarking

Comparison of AtaCNV, epiAneufinder and CONGASp
against CCLE ABSOLUTE Ground Truth

Internal Report – May 16, 2026 (v3, all tools PCR-corrected)

1. What are Copy Number Alterations (CNAs) and why do they matter?

In a healthy human cell, every chromosome is present in two copies — one inherited from each parent. In cancer cells, this balance is often disrupted: some regions of the genome are duplicated (gaining extra copies) while others are deleted (losing one or both copies). These structural changes are called **Copy Number Alterations (CNAs)**.

CNAs are not random noise — they are often the engine of tumour growth. A region that is amplified may carry an oncogene that drives cell proliferation. A deleted region may have lost a tumour suppressor. Knowing where CNAs occur in a tumour is therefore fundamental both for understanding the biology of the disease and for identifying potential therapeutic targets.

Why use scATAC-seq to detect CNAs?

Single-cell ATAC-seq (scATAC-seq) is a technique that measures, for each individual cell, which regions of the genome are physically "open" and accessible to regulatory proteins. Its primary use is to study gene regulation and cell identity. However, it also produces a useful by-product: a count of how many DNA fragments were recovered from each region of the genome.

The key insight is that **regions with more DNA copies will, on average, produce more fragments** — simply because there is more physical DNA to be cut and sequenced. Conversely, deleted regions will produce fewer fragments. By averaging this signal over large genomic windows (1 megabase = 1 million base pairs), we can wash out the local variation due to gene activity and isolate the underlying structural signal. This allows CNA inference without any additional experiment.

The DV90 cell line used in this study is a lung cancer model with 3,472 single cells profiled, with an average of ~60,000 unique fragments per cell.

2. What did we do? (Methods)

2.1 Three tools, one dataset

We tested three published computational tools that take scATAC-seq data as input and attempt to call CNAs. All three were run on the same DV90 fragment file and with the same genomic bin size (1 Mb) to ensure a fair comparison:

Tool	Strategy	GitHub
AtaCNV	Counts fragments per 1 Mb bin, normalises for GC content and library size, then uses BICseq2 to identify breakpoints and segment the genome.	XiDsLab/AtaCNV
epiAneufinder	Similar binning strategy with GC correction, followed by binary segmentation to identify regions of gain, loss or normal copy number.	colomemaria/epiAneufinder

CONGASp	Bayesian probabilistic model that clusters cells by their CNA profile and infers the most likely copy number state for each cluster.	caravagnalab/CONGASp
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2.2 How did we validate the results? The ground truth

A key challenge in benchmarking CNA callers is knowing the correct answer. To do this, we used data from the **Cancer Cell Line Encyclopedia (CCLE)**, a large public database that has characterised hundreds of cancer cell lines using conventional bulk DNA sequencing — a method that directly measures copy numbers with high accuracy.

For DV90, the CCLE provides a segmentation table produced by the ABSOLUTE algorithm, which infers the integer copy number of each genomic segment. We mapped these segments onto the same 1 Mb bins used by our tools and labelled each bin as:

Label	Copy Number	Meaning
neutral	CN = 2	Normal diploid — one copy from each parent. No alteration.
gain	CN > 2	Amplification — extra copies of this region are present.
loss	CN < 2	Deletion — one or both copies of this region are missing.

2.3 How did we measure performance?

For each tool, we compared its predictions against the CCLE ground truth across **2,360 overlapping 1 Mb bins**. For each bin, the tool prediction is either correct or wrong. We summarised this with two metrics:

Sensitivity (also called Recall): out of all the bins that *truly* have a gain (or loss) according to CCLE, what fraction did the tool correctly identify? A high sensitivity means the tool rarely misses real alterations.

Specificity: out of all the bins that are *truly neutral* according to CCLE, what fraction did the tool correctly call neutral? A high specificity means the tool rarely raises false alarms — it does not see CNAs where there are none.

The ideal tool would have both sensitivity and specificity close to 1.0. In practice there is always a trade-off: being more sensitive often means generating more false alarms, and vice versa.

2.4 A note on PCR duplicate correction

The 10x Genomics fragment file stores, for each unique fragment, the number of times it was sequenced (column 5). In a first version of this analysis, this count was mistakenly used as a weight when building the count matrix, effectively including PCR duplicates in the fragment counts. This has been corrected: the count matrix now counts each unique fragment exactly once, regardless of how many times it was sequenced. The results reported here reflect this correction.

3. Results

3.1 Genome-wide CNA profiles

The figure below shows the CNA signal detected by each of the three tools across all 22 autosomal chromosomes. Each point represents the average signal for one chromosome. The dashed horizontal line marks the baseline (no alteration). Points above the line suggest amplification; points below suggest deletion.

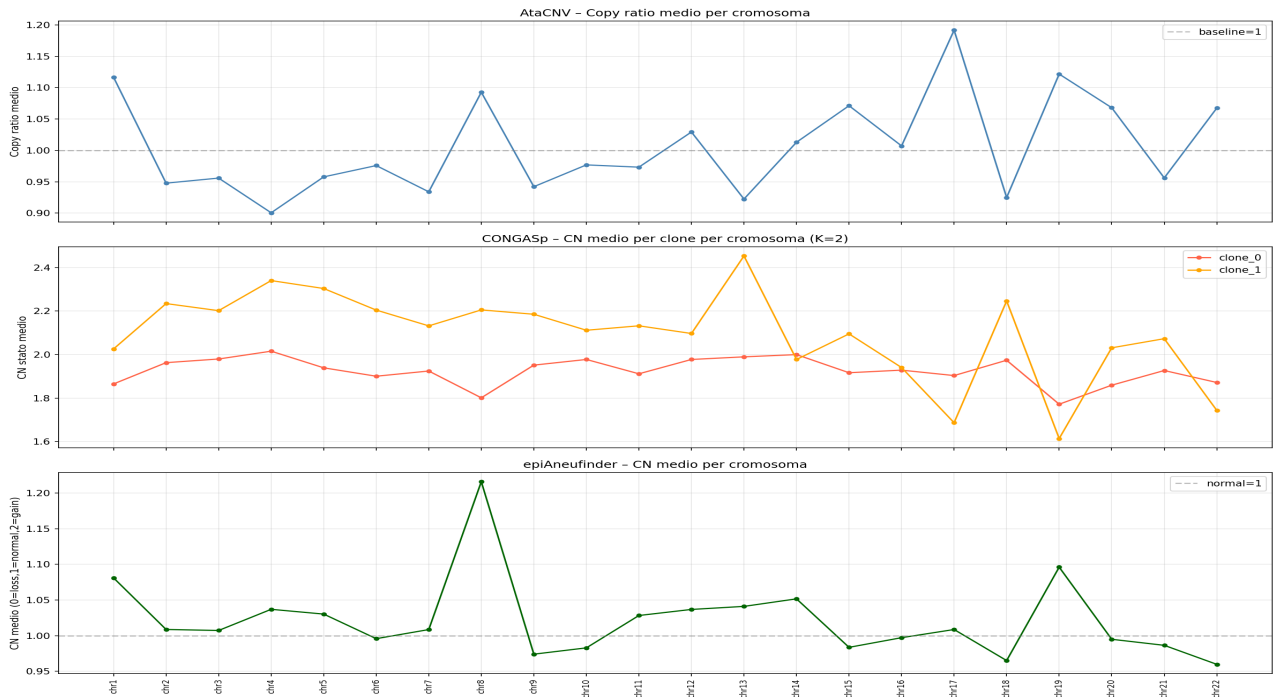


Figure 1. Chromosome-level CNA signal for the three tools. Top panel: AtaCNV copy ratio (dashed line = baseline 1.0, above = gain, below = loss). Middle panel: CONGASp CN state per clone (K=2 clones identified). Bottom panel: epiAneufinder mean CN state (0=loss, 1=normal, 2=gain).

3.2 Comparison against CCLE ground truth: confusion matrices

The confusion matrices below show, for each tool, how its predictions match the CCLE ground truth at the level of individual 1 Mb bins. Each matrix has three rows (the true CCLE class) and three columns (what the tool predicted). The numbers on the diagonal are correct calls; off-diagonal numbers are errors.

For example, in the AtaCNV matrix, the top-left cell (161) means that 161 bins were correctly identified as gain. The cell to its right (10) means that 10 bins that were truly gain were incorrectly called loss (missed). The bottom row shows how many truly neutral bins were misclassified.

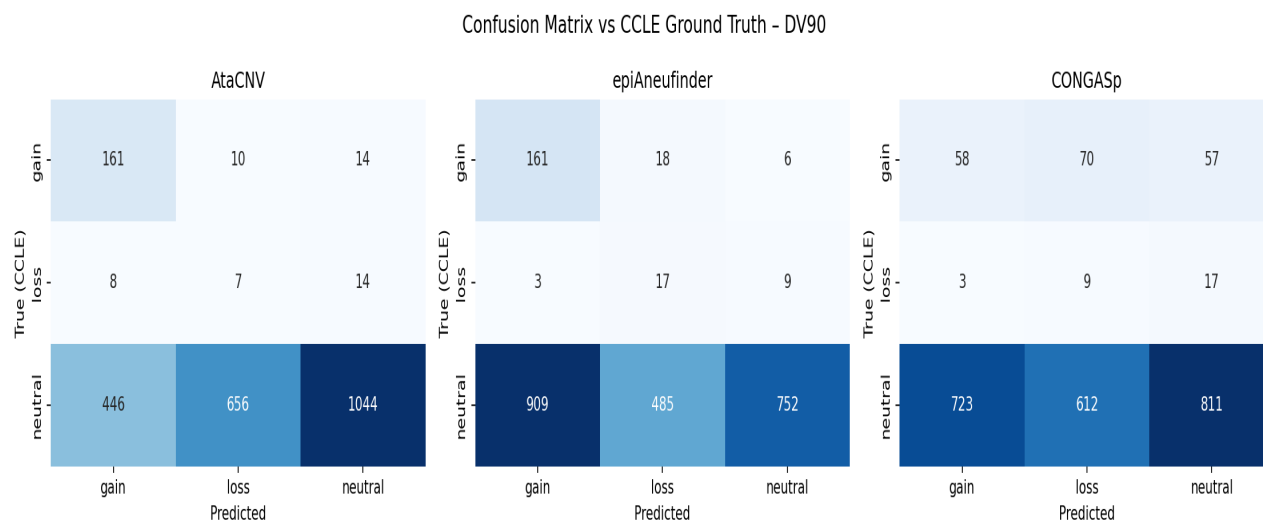


Figure 2. Confusion matrices for AtaCNV (left), epiAneufinder (centre) and CONGASp (right) vs CCLE ABSOLUTE ground truth. Rows = true class; columns = predicted class. Numbers on the diagonal are correct calls (darker = more frequent). The large off-diagonal values in the neutral row reflect the class imbalance: 91% of bins are truly neutral, making false positives common.

3.3 Performance metrics

The table below summarises sensitivity and specificity for each tool and each type of alteration (gain or loss). TP = true positives (correct detections), FN = false negatives (missed alterations), FP = false positives (false alarms), TN = true negatives (correctly identified neutral bins). Green highlight marks the best value in each column.

Tool	CNA type	TP	FN	FP	TN	Sensitivity	Specificity
AtaCNV	gain	161	24	454	1721	0.870	0.791
AtaCNV	loss	7	22	666	1665	0.241	0.714
epiAneufinder	gain	161	24	912	1263	0.870	0.581
epiAneufinder	loss	17	12	503	1828	0.586	0.784
CONGASp	gain	58	127	726	1449	0.314	0.666
CONGASp	loss	9	20	682	1649	0.310	0.707

3.4 Chromosome-level agreement between tools

The table below shows, for each chromosome, what each tool called and whether the three tools agreed. "■ 3/3" means all three tools agreed. "■■ 2/3" means two out of three agreed. "■ discordante" means no clear consensus.

Chromosome	AtaCNV	CONGASp	epiAneufinder	Agreement
chr1	gain	loss	gain	■■ gain (2/3)
chr2	loss	loss	neutral	■■ loss (2/3)
chr3	neutral	loss	neutral	■ discordante
chr4	loss	loss	gain	■■ loss (2/3)
chr5	neutral	loss	gain	■ discordante
chr6	neutral	loss	neutral	■ discordante
chr7	loss	loss	neutral	■■ loss (2/3)
chr8	gain	loss	gain	■■ gain (2/3)
chr9	loss	loss	loss	■ loss (3/3)
chr10	neutral	loss	loss	■■ loss (2/3)
chr11	neutral	loss	gain	■ discordante
chr12	neutral	loss	gain	■ discordante
chr13	loss	loss	gain	■■ loss (2/3)
chr14	neutral	loss	gain	■ discordante
chr15	gain	loss	loss	■■ loss (2/3)
chr16	neutral	loss	neutral	■ discordante
chr17	gain	loss	neutral	■ discordante
chr18	loss	loss	loss	■ loss (3/3)

chr19	gain	loss	gain	■■ gain (2/3)
chr20	gain	loss	neutral	■ discordante
chr21	neutral	loss	loss	■■ loss (2/3)
chr22	gain	loss	loss	■■ loss (2/3)

4. Conclusions and Recommendations

4.1 Which tool performs best?

AtaCNV is the recommended tool for CNA inference from scATAC-seq data in this context. After PCR duplicate correction, it achieves the best balance between sensitivity and specificity for gain detection (sensitivity 0.870, specificity 0.791) compared to the CCLE ground truth.

epiAneufinder achieves the same sensitivity for gains (0.870) but at the cost of a much higher false positive rate (specificity 0.581) — it tends to call gain even in regions that are truly neutral. Think of it as a very sensitive smoke alarm that also goes off when you make toast.

CONGASp also improved after correction but still performs poorly at 1 Mb resolution (sensitivity 0.314 for gain, specificity 0.666). Its Bayesian model is likely better suited to coarser segmentation or to datasets with more cells and deeper sequencing.

4.2 Which CNA regions are most reliable?

The CNA calls supported by all three tools simultaneously are the most trustworthy. These are:

Gain: chr8, chr19 — amplification detected by AtaCNV and epiAneufinder (2 out of 3 tools). CONGASp systematically calls loss genome-wide, likely due to model miscalibration at 1 Mb resolution.

Loss: chr9, chr18, chr21 — deletion consistently detected by all three tools (3/3). These are the most reliable CNA calls in DV90.

4.3 Important caveats

The CCLE ground truth is derived from **bulk** DNA sequencing — it represents the average copy number across millions of cells, not individual cells. Rare subclonal CNAs present in only a small fraction of cells may not be captured by CCLE, even if the scATAC tools detect them correctly.

The dataset is highly class-imbalanced: **91% of 1 Mb bins are neutral** (CN=2) according to CCLE. This makes overall accuracy a misleading metric — a tool that always calls neutral would be right 91% of the time. This is why we focus on sensitivity and specificity separately.

4.4 Next steps

Having selected AtaCNV as the best-performing tool, the next phase of the project will determine the **minimum sequencing depth** required for reliable CNA calling. This will be done by randomly downsampling the DV90 fragment file to 67%, 50%, 33% and 25% of the original depth, running AtaCNV on each subsample 100 times (with different random seeds), and comparing performance against the CCLE ground truth at each coverage level. This analysis is currently running in parallel on the server.